



DR ESLAM AHMED
eliteigcse.com

3D Pythagoras and Trigonometry

Every 3D question is two 2D triangles: one flat on the base, one standing up. Draw each one separately.

THE TOPIC ON ONE MAP

3D Pythagoras and Trigonometry

01 — Lengths in 3D

space diagonal $\sqrt{a^2 + b^2 + c^2}$ does the pencil fit?

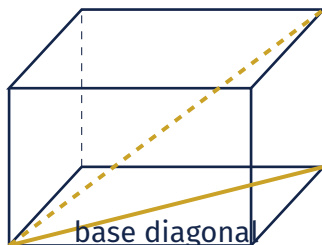
02 — Angle between a line and a plane

drop a perpendicular base triangle, then tan

03 — Harder 3D problems

prisms and pyramids midpoints exact form

TWO TRIANGLES Base first, then the upright triangle:



- 1 base triangle: Pythagoras on the floor
- 2 upright triangle: base diagonal + height
- 3 angle with the base: $\tan \theta = \frac{\text{height}}{\text{base diagonal}}$

01 Lengths in 3D

Pythagoras twice: once across the base, once up to the top — or all at once with $d = \sqrt{a^2 + b^2 + c^2}$.

WHEN YOU SEE → YOU DO

“length of the diagonal AG ” of a cuboid → $\sqrt{a^2 + b^2 + c^2}$

“does the pencil fit?” → compare with the space diagonal

a point at a midpoint → write its three distances, then Pythagoras

“in the form $\sqrt{a}$ ” → do not round

WORKED MODEL • JUNE 2022 P1HR

Cube of side 6; T is the midpoint of AB , V is the midpoint of CH . Find TV .

1 differences: 3 across, 6 back, 3 down

three distances

ANSWER $\sqrt{3^2 + 6^2 + 3^2} = \sqrt{54}$ cm

IDEA 01

Practise: Lengths in 3D

My questions first, then selected practice, then real past papers.

Q1 **DR ESLAM** Space diagonalCH 4 • TOPIC 12 **2 marks**

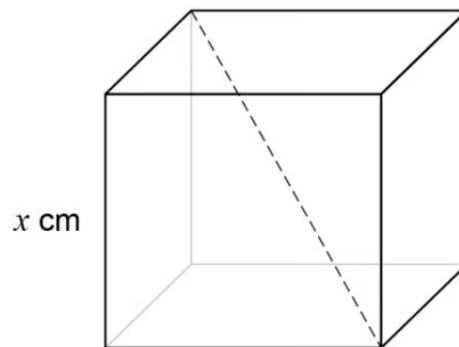
A cuboid is 3 cm by 4 cm by 12 cm. Find the length of its space diagonal.

SELECTED PRACTICE

Q2 PRACTICE Practice · Medium

CH 4 · TOPIC 12 1 mark

- 6 Here is a cube with edge length x cm
One diagonal is shown.



Circle the length, in centimetres, of the diagonal.

$\sqrt{3} x$	$\sqrt[3]{3x^2}$	$\sqrt{x^3}$	$\sqrt[3]{3} x$
--------------	------------------	--------------	-----------------

(1 mark)

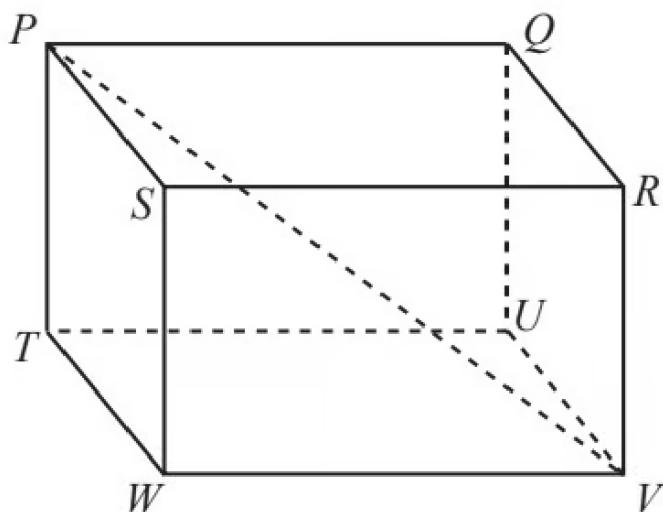
Q3

PRACTICE

Practice · Medium

CH 4 · TOPIC 12 3 marks

9



NOT TO
SCALE

The diagram shows a cuboid $PQRSTU VW$. $PV = 17.2$ cm

The angle between the line PV and the base $TUVW$ of the cuboid is 43° .

Calculate PT .

$PT = \dots\dots\dots$ cm

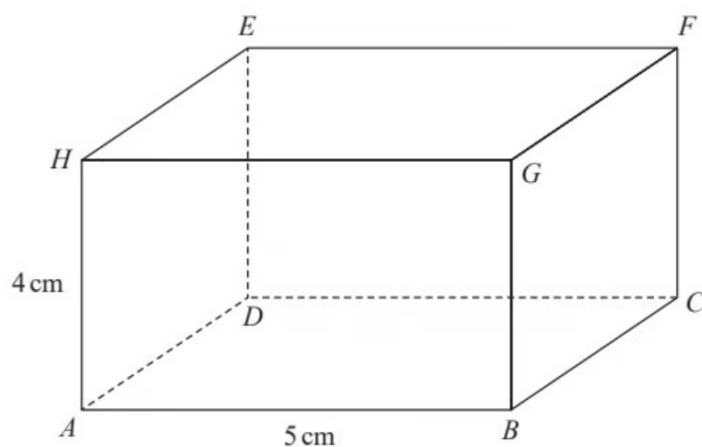
(3 marks)

Q4**PRACTICE**

Practice · Hard

CH 4 · TOPIC 12 5 marks

4 The diagram shows cuboid

Diagram **NOT**
accurately drawn

$$AB = 5 \text{ cm}$$

$$AH = 4 \text{ cm}$$

The size of the angle between CH and the plane $ABCD$ is 35°

Calculate the volume of the cuboid.

Give your answer correct to 3 significant figures.

.....cm³

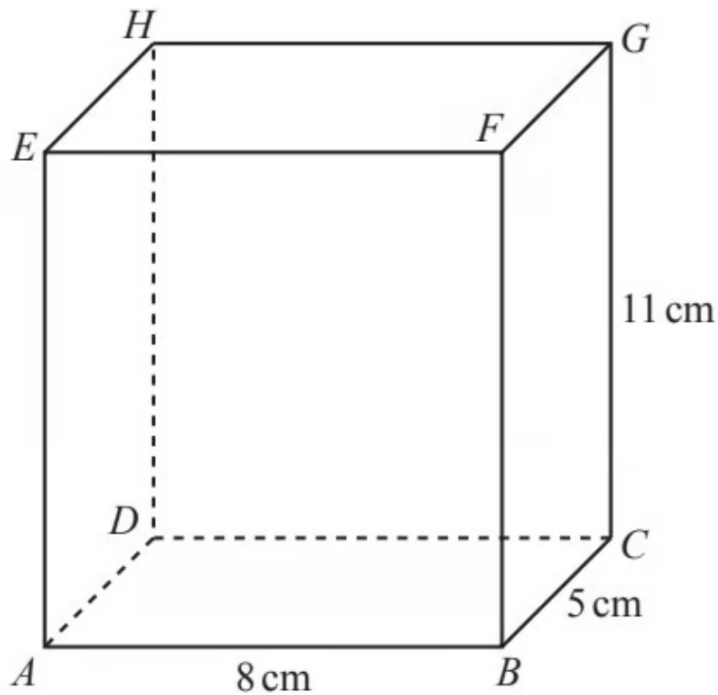
(5 marks)

Q5**PRACTICE**

Practice · Very Hard

CH 4 · TOPIC 12 4 marks

10

NOT TO
SCALE

$ABCDEFGH$ is a cuboid. $AB = 8$ cm, $BC = 5$ cm and $CG = 11$ cm.

Ivana has a pencil of length 13 cm.

Does this pencil fit completely inside the cuboid? Show how you decide.

(4 marks)

PAST PAPERS

Q6 PAST PAPER Edexcel 4MA1 · June 2022 · Paper 1HR · Q18

CH 4 · TOPIC 12 4 marks

18 The diagram shows a cube $ABCDEFGH$ with sides of length 6 cm.

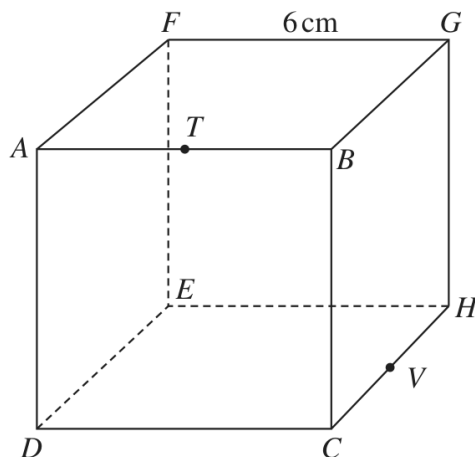


Diagram **NOT**
accurately drawn

T is the midpoint of AB and V is the midpoint of CH

Work out the distance from T to V in a straight line through the cube.

Give your answer in the form $\sqrt{a}$ cm where a is an integer.

..... cm

(Total for Question 18 is 4 marks)

Q7

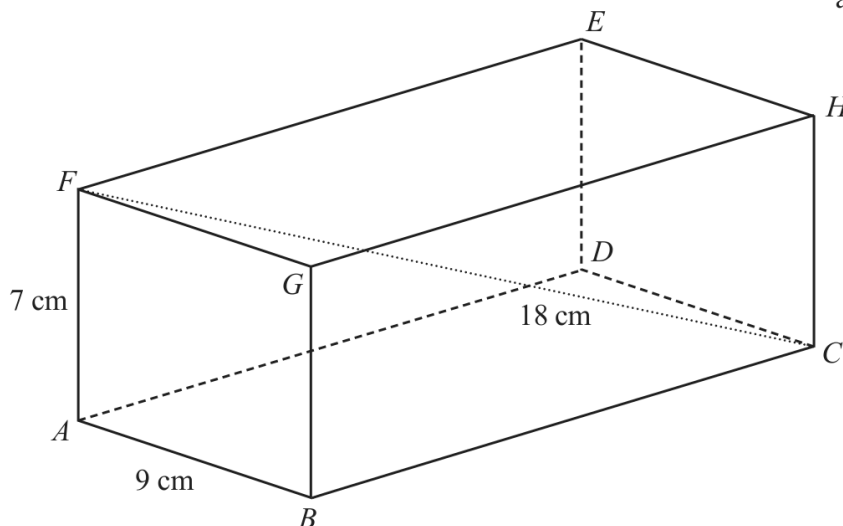
PAST PAPER

Edexcel 4MA1 · June 2025 · Paper 1HR · Q20

CH 4 · TOPIC 12 3 marks

20 The diagram shows cuboid $ABCDEFGH$

Diagram **NOT**
accurately drawn



$$AB = 9 \text{ cm} \quad AF = 7 \text{ cm} \quad FC = 18 \text{ cm}$$

Calculate the length of BC

Give your answer correct to 3 significant figures.

..... cm

(Total for Question 20 is 3 marks)

26 Here is a prism $ABCDEFGH$ with a horizontal, rectangular base $ABGF$

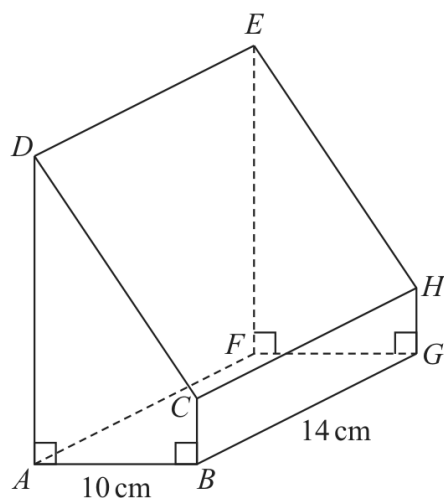


Diagram **NOT**
accurately drawn

$$AB = 10\text{ cm} \qquad BG = 14\text{ cm}$$

$$\text{angle } DAB = \text{angle } ABC = \text{angle } EFG = \text{angle } FGH = 90^\circ$$

$$BC = \frac{1}{5}AD$$

The angle of elevation of E from B is 50°

Calculate the volume of the prism.

Give your answer correct to 3 significant figures.

Show your working clearly.

.....cm³

(Total for Question 26 is 5 marks)

02 The Angle Between a Line and a Plane

From the top end of the line, drop straight down onto the plane. The angle is at the bottom end, in the right-angled triangle you have just made.

WHEN YOU SEE → YOU DO

“angle between AH and the plane $EFGH$ ”

→ drop to the plane, find the base length

a pyramid, “angle between VA and the base”

→ half the base diagonal, then tan

a flagpole on the ground

→ horizontal distance first

“angle between a sloping edge and the base is 55° ”

→ work backwards for the height

Drop a perpendicular

Base length (Pythagoras)

$\tan = \text{height} \div \text{base}$

WORKED MODEL • JANUARY 2020 P1H

Cuboid: $EH = 9$, $HG = 5$, $GB = 6$. Angle between AH and $EFGH$?

1 A is above F: $FH = \sqrt{9^2 + 5^2} = \sqrt{106}$

base triangle

2 $\tan \theta = \frac{6}{\sqrt{106}}$

upright triangle

ANSWER $\theta = 30.2^\circ$

× **TRAP** Using the sloping line's length as the base. The base of the upright triangle lies flat on the plane.

IDEA 02

Practise: The Angle Between a Line and a Plane

My questions first, then selected practice, then real past papers.

Q9 DR ESLAM Line and plane

CH 4 • TOPIC 12 3 marks

A cuboid has base 6 cm by 8 cm and height 5 cm. Find the angle between a space diagonal and the base.

Q10 DR ESLAM Pyramid

CH 4 • TOPIC 12 3 marks

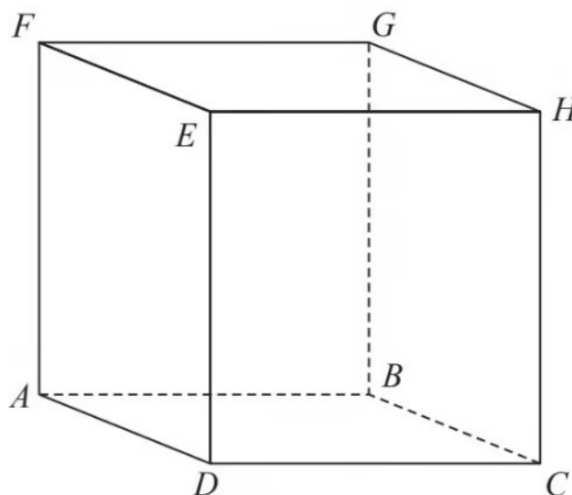
A square-based pyramid has base 10 cm and vertical height 12 cm. Find the angle between a sloping edge and the base.

SELECTED PRACTICE

Q11 PRACTICE Practice • Medium

CH 4 • TOPIC 12 4 marks

- 1 $ABCDEFGH$ is a cuboid.



$$AB = 7.3 \text{ cm}$$

$$CH = 8.1 \text{ cm}$$

$$\text{Angle } BCA = 48^\circ$$

Find the size of the angle between AH and the plane $ABCD$.
Give your answer correct to 1 decimal place.

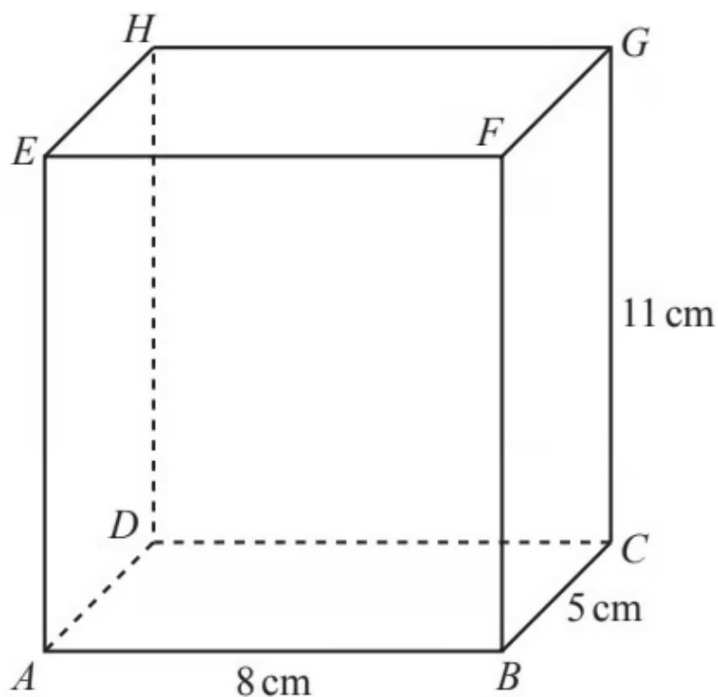
(4 marks)

Q12**PRACTICE**

Practice · Medium

CH 4 · TOPIC 12 4 marks

10

NOT TO
SCALE

$ABCDEFGH$ is a cuboid. $AB = 8$ cm, $BC = 5$ cm and $CG = 11$ cm.

i) Calculate angle CAB .

Angle $CAB = \dots\dots\dots$ [2]

ii) Calculate angle GAC .

Angle $GAC = \dots\dots\dots$ [2]

(4 marks)

Q13**PRACTICE**

Practice • Hard

CH 4 • TOPIC 12 4 marks

2 Here is a cube $ABCDEFGH$.

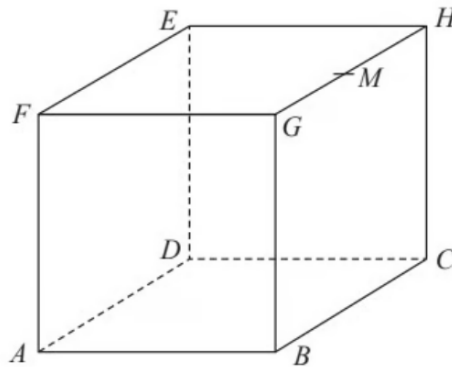


Diagram **NOT**
accurately drawn

M is the midpoint of the edge GH .

Find the size of the angle between the line MA and the plane $ABCD$.

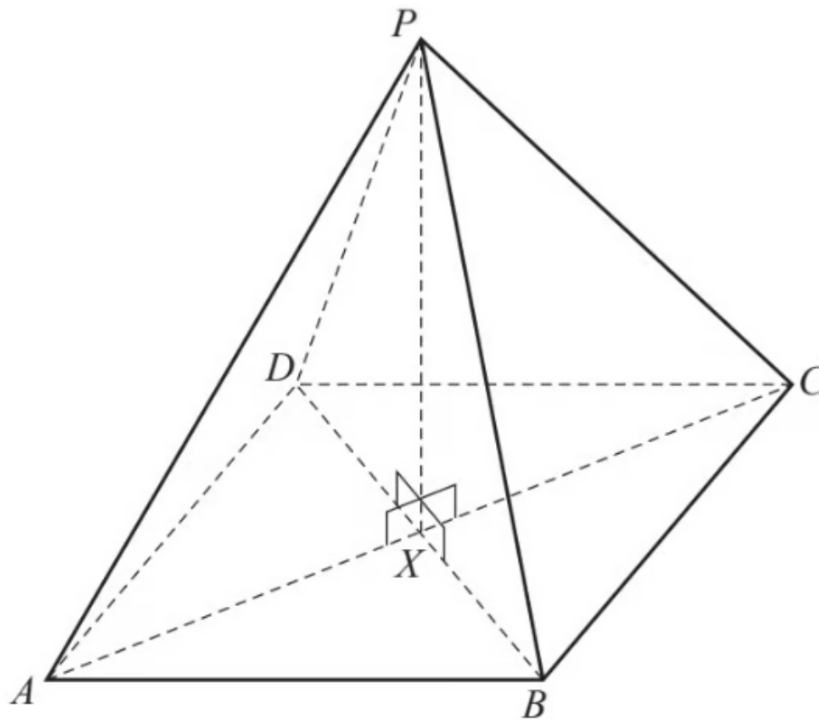
Give your answer correct to 1 decimal place.

(4 marks)

Q14 PRACTICE Practice • Very Hard

CH 4 • TOPIC 12 3 marks

11

NOT TO
SCALE

The diagram shows a pyramid with a square base $ABCD$. $DB = 8$ cm.
 P is vertically above the centre, X , of the base and $PX = 5$ cm.

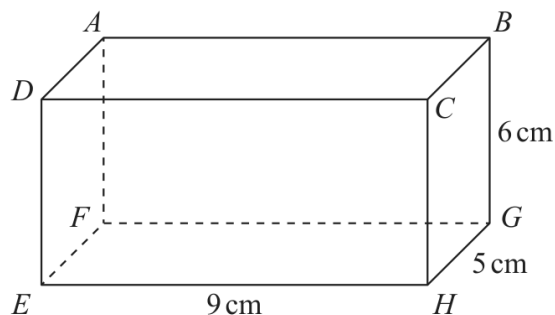
Calculate the angle between PB and the base $ABCD$.

(3 marks)

PAST PAPERS

Q15 PAST PAPER Edexcel 4MA1 · January 2020 · Paper 1H · Q19

CH 4 · TOPIC 12 4 marks

19 The diagram shows a cuboid $ABCDEFGH$.Diagram **NOT**
accurately drawn $EH = 9\text{ cm}$, $HG = 5\text{ cm}$ and $GB = 6\text{ cm}$.

Work out the size of the angle between AH and the plane $EFGH$.
Give your answer correct to 3 significant figures.

o

(Total for Question 19 is 4 marks)

Q16**PAST PAPER**

Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q17

CH 4 • TOPIC 12 6 marks

17 The diagram shows a prism $ABCDEFGH$ with a horizontal base.

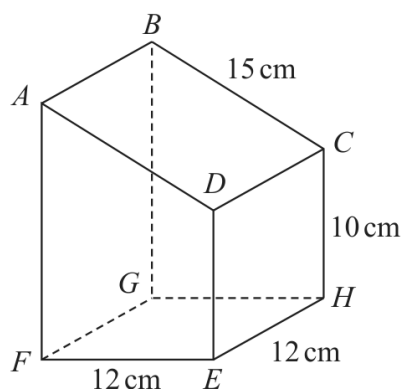


Diagram **NOT**
accurately drawn

The base of the prism, $EFGH$, is a square of side 12 cm.

Trapezium $ADEF$ is a cross section of the prism where AF and DE are vertical edges.

$$DE = CH = 10 \text{ cm}$$

$$AD = BC = 15 \text{ cm}$$

- (a) Work out the size of the angle between CF and the base $EFGH$.

Give your answer correct to one decimal place.

..... °

- (b) Work out the length of BE .

Give your answer correct to one decimal place.

..... cm

(Total for Question 17 is 6 marks)

Q17 PAST PAPER Edexcel 4MA1 · May 2021 · Paper 1H · Q22

CH 4 · TOPIC 12 4 marks

22 The diagram shows a triangular prism $ABCDEF$ with a horizontal base $ABEF$.

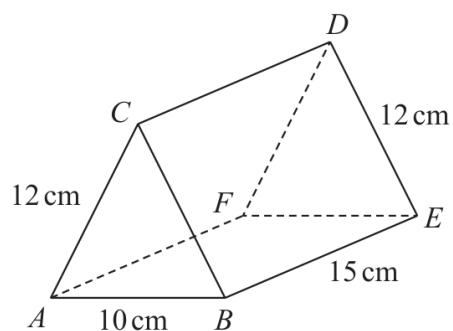


Diagram **NOT**
accurately drawn

$$AC = BC = FD = ED = 12 \text{ cm} \quad AB = 10 \text{ cm} \quad BE = 15 \text{ cm}$$

Calculate the size of the angle between AD and the base $ABEF$.
Give your answer correct to 3 significant figures.

.....°

(Total for Question 22 is 4 marks)

Q18**PAST PAPER**

Edexcel 4MA1 · January 2022 · Paper 2HR · Q19

CH 4 · TOPIC 12 3 marks

19 $ABCD$ is a horizontal rectangular field.

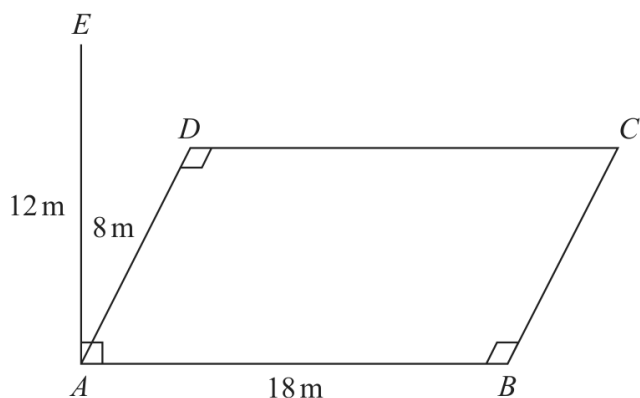


Diagram **NOT**
accurately drawn

A vertical pole, AE , is placed at the corner A of the field.

$$AE = 12 \text{ m} \quad AB = 18 \text{ m} \quad AD = 8 \text{ m}$$

Calculate the size of the angle between EC and the plane $ABCD$
Give your answer correct to one decimal place.

○

(Total for Question 19 is 3 marks)

Q19**PAST PAPER**

Edexcel 4MA1 · June 2025 · Paper 2H · Q24

CH 4 · TOPIC 12 4 marks

24 The diagram shows a square-based pyramid $ABCDE$

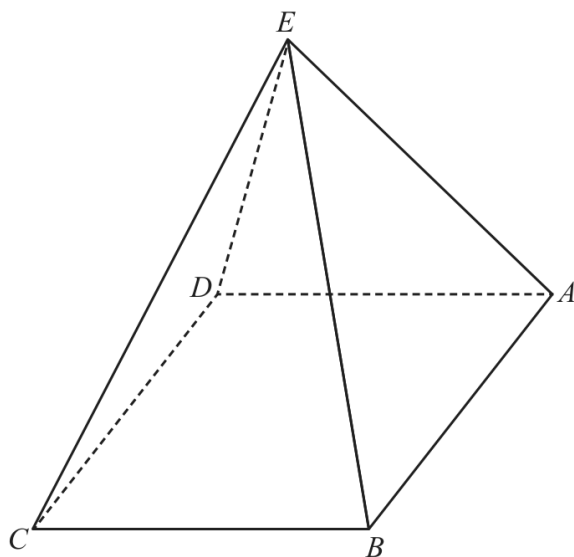


Diagram **NOT**
accurately drawn

$$EA = EB = EC = ED$$

M is the centre of the horizontal square base $ABCD$

Q is the midpoint of AB

Angle $EQM = 80^\circ$

$$EA : AB = n : 1$$

Find the value of n

Give your answer correct to 3 significant figures.

$$n = \dots\dots\dots$$

(Total for Question 24 is 4 marks)

03 Harder 3D Problems

Set the shape on coordinates in your head: which lengths are on the floor, which are vertical. Solve one triangle at a time, keeping exact values.

WHEN YOU SEE → YOU DO

a prism with a sloping face

→ the height comes from the cross-section

a trapezium cross-section

→ overhang = $\frac{1}{2}$ (difference of the parallel sides)

“give T in the form $\frac{\sqrt{p}+\sqrt{q}}{17}$ ”

→ exact surds, rationalise at the end

a midpoint of a base or edge

→ halve the coordinates

WORKED MODEL • JUNE 2025 P2HR

Rectangular base, $AC = 40$ (diagonal), E is above C , angle $CAE = 35^\circ$, M the midpoint of AC . Angle CME ?

1 $CE = 40 \tan 35^\circ$

triangle ACE

2 $\tan CME = \frac{40 \tan 35^\circ}{20}$

triangle MCE

ANSWER $CME = 54.5^\circ$

IDEA 03

Practise: Harder 3D Problems

My questions first, then selected practice, then real past papers.

SELECTED PRACTICE

Q20**PRACTICE**

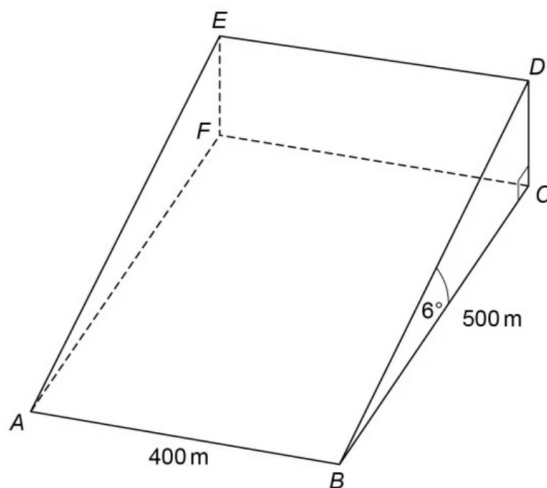
Practice • Hard

CH 4 • TOPIC 12 6 marks

7 (a) $ABCDEF$ is a triangular prism which represents part of a hill.

$ABCF$ is the horizontal rectangular base.

D is vertically above C .

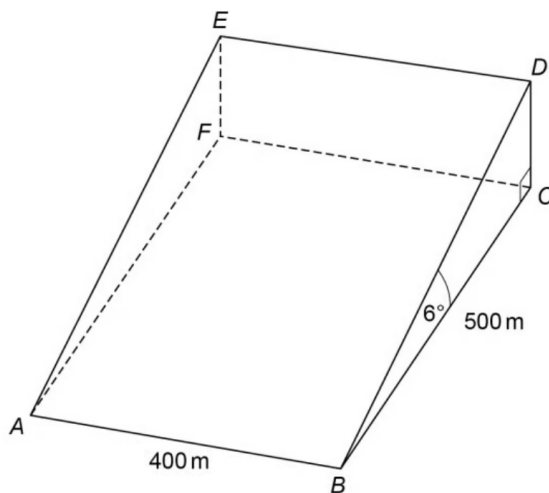


Work out the height CD .

.....m

(2 marks)

(b) Jamil walks in a straight line from A to D .



Work out the size of angle DAC .

You **must** show your working.

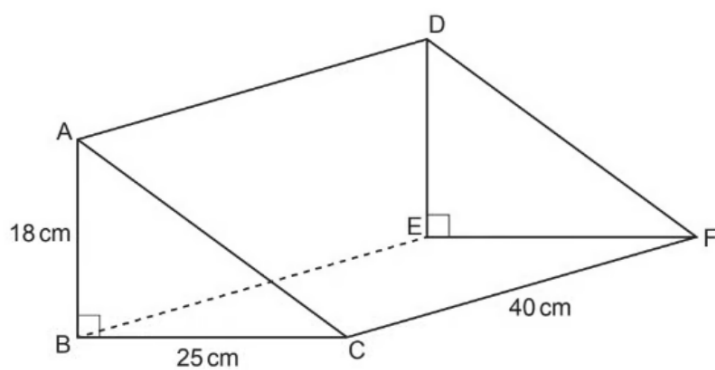
.....degrees

(4 marks)

Q21 PRACTICE Practice · Hard

CH 4 · TOPIC 12 6 marks

12 The diagram shows a right-angled triangular prism ABCDEF.



Calculate angle AFB.

(6 marks)

Q22 PRACTICE Practice · Very Hard

CH 4 · TOPIC 12 3 marks

- 4 The diagram shows a solid prism $ABCDEFGH$.

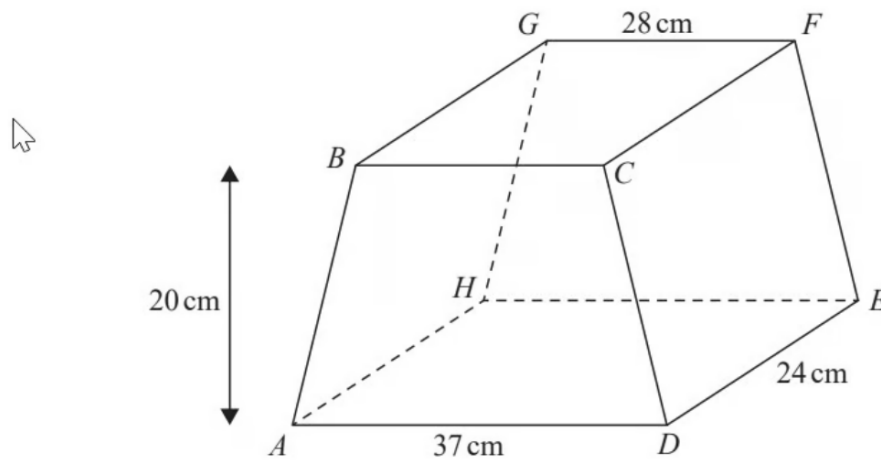


Diagram **NOT**
accurately drawn

The trapezium $ABCD$, in which AD is parallel to BC , is a cross section of the prism. The base $ADEH$ of the prism is a horizontal plane. $ADEH$ and $BCFG$ are rectangles. The midpoint of BC is vertically above the midpoint of AD so that $BA = CD$.

$$AD = 37 \text{ cm} \quad GF = 28 \text{ cm} \quad DE = 24 \text{ cm}$$

The perpendicular distance between edges AD and BC is 20 cm.

Calculate the size of the angle between AF and the plane $ADEH$.
Give your answer correct to one decimal place.

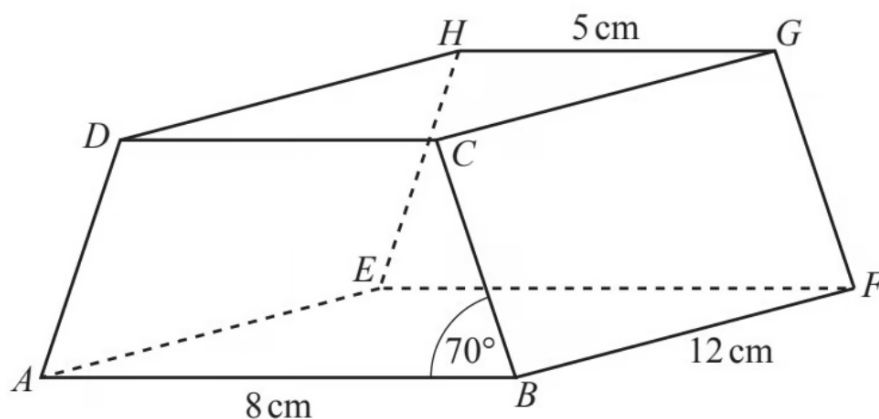
(3 marks)

Q23**PRACTICE**

Practice • Very Hard

CH 4 • TOPIC 12 5 marks

9

NOT TO
SCALE

The diagram shows a prism with a rectangular base, $ABFE$.

The cross-section, $ABCD$, is a trapezium with $AD = BC$. $AB = 8$ cm, $GH = 5$ cm, $BF = 12$ cm and angle $ABC = 70^\circ$.

The perpendicular from G onto EF meets EF at X .

i) Show that $EX = 6.5$ cm.

[1]

ii) Calculate AX .

$AX = \dots\dots\dots$ cm [2]

iii) Calculate the angle between the diagonal AG and the base $ABFE$.

[2]

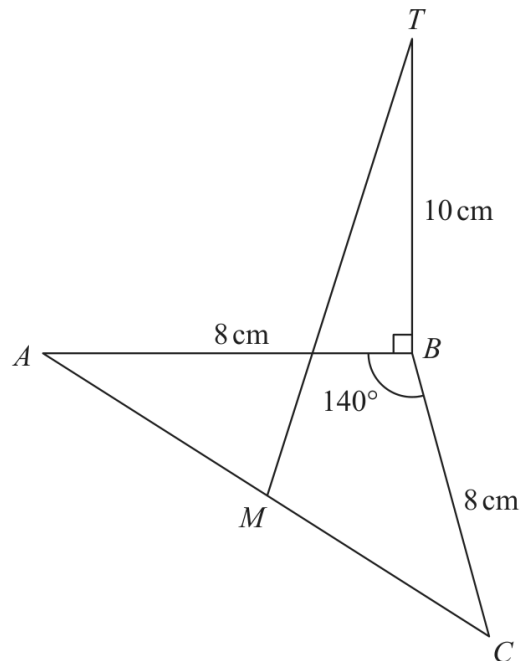
(5 marks)

PAST PAPERS

Q24 PAST PAPER Edexcel 4MA1 · January 2021 · Paper 2HR · Q22

CH 4 · TOPIC 12 4 marks

- 22 ABC is an isosceles triangle in a horizontal plane.
The point T is vertically above B .

Diagram **NOT**
accurately drawnAngle $ABC = 140^\circ$ $AB = BC = 8 \text{ cm}$ $TB = 10 \text{ cm}$ M is the midpoint of AC .

Calculate the size of the angle between MT and the horizontal plane ABC .
Give your answer correct to one decimal place.

o

(Total for Question 22 is 4 marks)

Q25**PAST PAPER**

Edexcel 4MA1 · January 2022 · Paper 1H · Q21

CH 4 · TOPIC 12 5 marks

21 The diagram shows the prism $ABCDEFGHJK$ with horizontal base $AEFG$

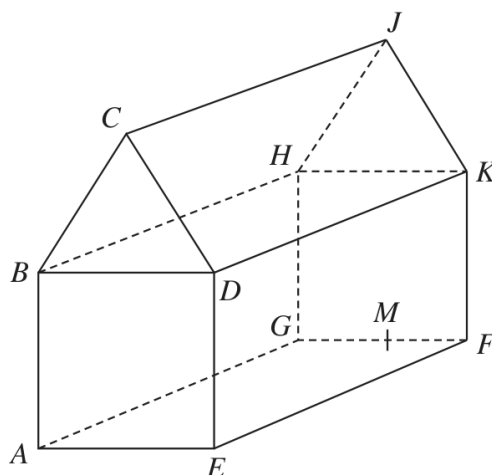


Diagram **NOT**
accurately drawn

$ABCDE$ is a cross section of the prism where
 $ABDE$ is a square
 BCD is an equilateral triangle

$$EF = 2 \times AE$$

M is the midpoint of GF so that JM is vertical.

Angle $MAJ = y^\circ$

Given that $\tan y^\circ = T$

find the value of T , giving your answer in the form $\frac{\sqrt{p} + \sqrt{q}}{17}$ where p and q are integers.

$T = \dots\dots\dots$

(Total for Question 21 is 5 marks)

Q26

PAST PAPER

Edexcel 4MA1 · June 2023 · Paper 2HR · Q22

CH 4 · TOPIC 12 **5 marks**

22 Here is a cuboid $ABCDEFGH$

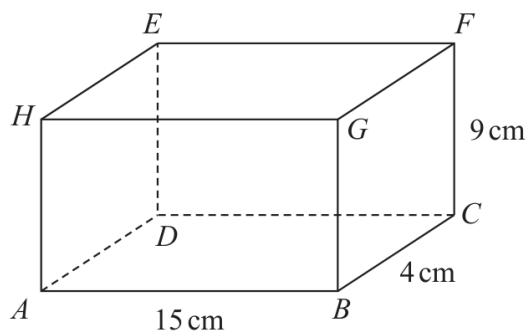


Diagram **NOT**
accurately drawn

$$AB = 15 \text{ cm} \quad BC = 4 \text{ cm} \quad CF = 9 \text{ cm}$$

- (a) Work out the length of BE
Give your answer correct to 3 significant figures.

..... cm

Here is a cuboid $PQRSTUVW$

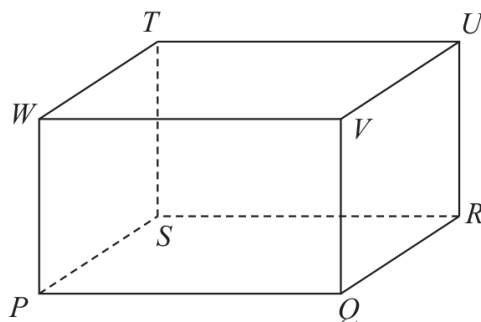


Diagram **NOT**
accurately drawn

$$PR = 42 \text{ cm}$$

The size of the angle between PU and the plane $PQRS$ is 30°

M is the midpoint of PR

- (b) Work out the size of angle UMR
Give your answer correct to 3 significant figures.

.....^o

(Total for Question 22 is 5 marks)

Q27

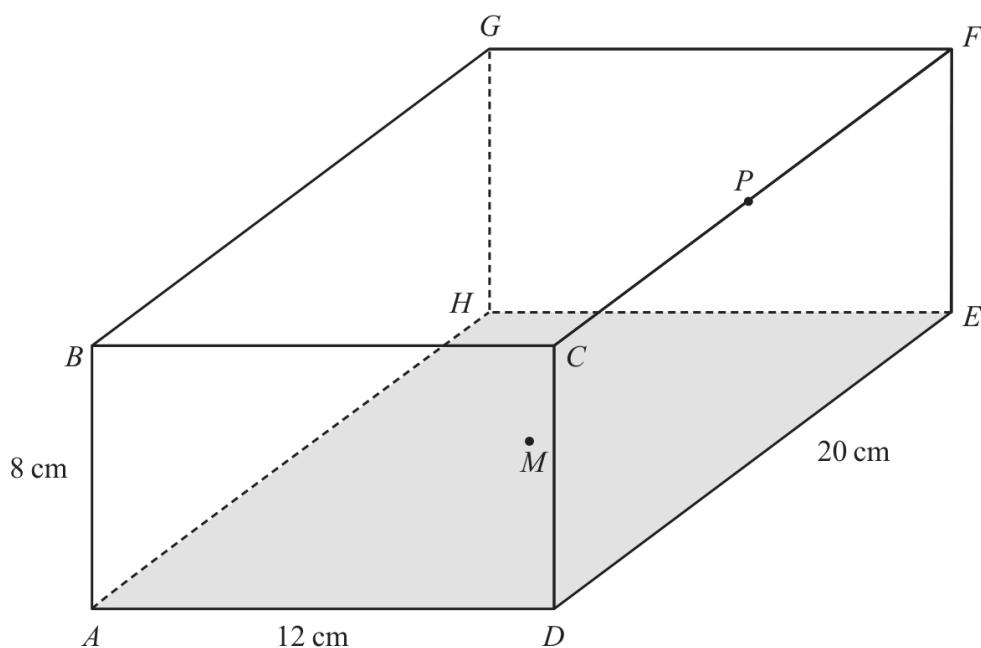
PAST PAPER

Edexcel 4MA1 · May 2024 · Paper 1HR · Q22

CH 4 · TOPIC 12 6 marks

22 The diagram shows a cuboid $ABCDEFGH$ with horizontal base $ADEH$

Diagram **NOT**
accurately drawn



$AB = 8 \text{ cm}$ $AD = 12 \text{ cm}$ $DE = 20 \text{ cm}$

M is the midpoint of the base $ADEH$ and P is the midpoint of the edge CF

Work out the size of angle BMP

Give your answer correct to one decimal place.

(Total for Question 22 is 6 marks)

Q28**PAST PAPER**

Edexcel 4MA1 · June 2025 · Paper 2HR · Q23

CH 4 · TOPIC 12 3 marks

- 23 The diagram shows a triangular prism, $ABCDEF$, with a horizontal rectangular base $ABCD$

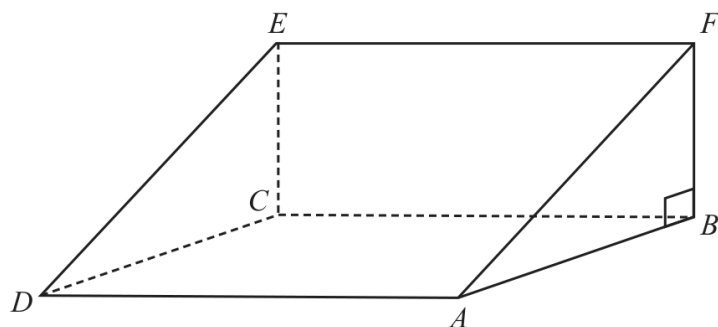


Diagram **NOT**
accurately drawn

M is the midpoint of the line AC

$AC = 40 \text{ cm}$ angle $CAE = 35^\circ$ angle $ABF = 90^\circ$

Work out the size of angle CME

Give your answer correct to 3 significant figures.

○

(Total for Question 23 is 3 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|-----------------------|--|
| 01 | Lengths | $\sqrt{a^2 + b^2 + c^2}$; midpoints: halve. |
| 02 | Line and plane | Drop a perpendicular; base by Pythagoras; tan. |
| 03 | Harder | One triangle at a time; exact values. |

BOTTOM LINE Floor triangle first → then the upright triangle → angle at the bottom.

Answers

3D Pythagoras and Trigonometry • for the teacher. Every answer checked with a computer algebra system.

IDEA 01 • LENGTHS IN 3D

- | | | |
|-----------|---|---|
| Q1 | Dr Eslam • Space diagonal | $\sqrt{9 + 16 + 144} = 13 \text{ cm}$ |
| Q2 | Practice • Medium | $\sqrt{3} x$ |
| Q3 | Practice • Medium | $PT = 17.2 \sin 43^\circ = 11.7 \text{ cm}$ |
| Q4 | Practice • Hard | $AC = \frac{4}{\tan 35^\circ} = 5.71, BC = 2.76: 55.3 \text{ cm}^3$ |
| Q5 | Practice • Very Hard | $\sqrt{64 + 25 + 121} = 14.5 \text{ cm} > 13: \text{yes, it fits}$ |
| Q6 | Edexcel 4MA1 • June 2022 • Paper 1HR • Q18 | $\sqrt{3^2 + 6^2 + 3^2} = \sqrt{54} \text{ cm}$ |
| Q7 | Edexcel 4MA1 • June 2025 • Paper 1HR • Q20 | $BC = \sqrt{324 - 81 - 49} = 13.9 \text{ cm}$ |
| Q8 | Edexcel 4MA1 • November 2025 • Paper 1H • Q26 | $BF = \sqrt{296}, AD = 20.5, BC = 4.10: 1720 \text{ cm}^3$ |

IDEA 02 • THE ANGLE BETWEEN A LINE AND A PLANE

- Q9** Dr Eslam • Line and plane
base diagonal 10: $\tan^{-1} \frac{5}{10} = 26.6^\circ$
- Q10** Dr Eslam • Pyramid
half diagonal $5\sqrt{2}$: $\tan^{-1} \frac{12}{7.07} = 59.5^\circ$
- Q11** Practice • Medium
 $BC = 6.57$: 50.9°
- Q12** Practice • Medium
(i) 32.0° (ii) 49.4°
- Q13** Practice • Hard
(cube side 2) $\tan^{-1} \frac{2}{\sqrt{5}} = 41.8^\circ$
- Q14** Practice • Very Hard
 $XB = 4$: $\tan^{-1} \frac{5}{4} = 51.3^\circ$
- Q15** Edexcel 4MA1 • January 2020 • Paper 1H • Q19
 30.2°
- Q16** Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q17
(a) 30.5° (b) $BE = 25.5$ cm
- Q17** Edexcel 4MA1 • May 2021 • Paper 1H • Q22
height 10.91, $AD = 19.2$: 34.6°
- Q18** Edexcel 4MA1 • January 2022 • Paper 2HR • Q19
 $AC = 19.7$: 31.4°
- Q19** Edexcel 4MA1 • June 2025 • Paper 2H • Q24
 $n = 2.92$

IDEA 03 • HARDER 3D PROBLEMS

- Q20** Practice · Hard (a) $CD = 500 \sin 6^\circ = 52.3 \text{ m}$ (b) 4.7°
- Q21** Practice · Hard $BF = \sqrt{25^2 + 40^2} = 47.2$: 20.9°
- Q22** Practice · Very Hard 26.3°
- Q23** Practice · Very Hard (i) $1.5 + 5 = 6.5$ (ii) $AX = 13.6 \text{ cm}$ (iii) height 4.12 : 16.8°
- Q24** Edexcel 4MA1 · January 2021 · Paper 2HR · Q22 $BM = 8 \cos 70^\circ = 2.74$: 74.7°
- Q25** Edexcel 4MA1 · January 2022 · Paper 1H · Q21 $T = \frac{2 + \sqrt{3}}{\sqrt{17}} = \frac{\sqrt{68} + \sqrt{51}}{17}$
- Q26** Edexcel 4MA1 · June 2023 · Paper 2HR · Q22 (a) $BE = 17.5 \text{ cm}$ (b) $UR = 42 \tan 30^\circ$: 49.1°
- Q27** Edexcel 4MA1 · May 2024 · Paper 1HR · Q22 78.6°
- Q28** Edexcel 4MA1 · June 2025 · Paper 2HR · Q23 $CE = 40 \tan 35^\circ$, $CM = 20$: 54.5°